# A Debate of Security vs Privacy and the Patriot Act

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# Report

## Executive Summary

With the ever-growing development of the internet and the cybersecurity landscape, one debate stands paramount to the rest in modern day digital societies – the debate between privacy and security. This research report navigates through the evolution of these concerns through pivotal moments in history, unraveling the intricate dance between an individual's cherished right to privacy and the dire societal need for everyday security. From the covert art-forms practiced during the Cold War era to a new digital society being born in the Early Internet era, a balance in the privacy and security debate was slowly but surely being discovered until the events of 9/11. This report delves into the effects of 9/11 on digital privacy and security, and how government workers such as Edward Snowden were able to turn the tides on the unbalanced government act between security and privacy. As this report traverses through time, it will uncover the dynamic landscape of societal and government action upon the debate, along with public perceptions and current day solutions to the matter. The report then finalizes on the modern day interplay between privacy and security, and what a present day individual can do in order to stay both secure and private while navigating the internet.

## Introduction

Privacy– a basic human right which advocates for safeguarding one's personal information and autonomy, is often at odds with that of security, society's greatest defender against the endless threats towards its people. Within this concept of privacy vs. security, there lies a fundamental truth: by enhancing one side of either privacy or security, compromises must be made on the other side for success. As the years went by, the pendulum of concerns between the two sides continuously swung back and forth within society, leading to opportunities for capitalization. The United States government, corporate business, and professional hackers all rose to the scene on this technological landscape, using people's fears and desire for simplicity to gather even more information from these individuals. This report will go over just how the government takes advantage of their security powers to invade citizen's privacy, how large corporations take advantage of privacy for the sake of "convenience" and profitability, and how hackers continuously infiltrate online security measures.

Overall, the continuously changing public perception of privacy and security is exactly the catalyst that dictates what is being taken away from the average digital user; their cybersecurity or their cyberprivacy. In order to fully understand the progression of this debate and how to effectively address the issue in 2023, one must first look at the beginnings of this modern-day issue, which began even before the cyber world that exists today.

# United States Before the Patriot Act

The Cold War era, which spanned from approximately 1947-1991, was the beginning stages of societies first developing major debates between the ideas of privacy and security from the United States government. During the Cold War, espionage was raised to an art form by both the United States and the Soviet Union. With the immense technological advancements during this time period, 3 major forms of intelligence collection were established by both sides. These were Human Intelligence (HUMINT), Signals Intelligence (SIGINT), and Photographic Intelligence (PHOTINT).

Human Intelligence, essentially "Spy vs. Spy", is one of the most basic forms of intelligence gathering where information is gathered through interpersonal interactions, infiltrations, and espionage [1]. The intricacies of this form of intelligence became pronounced during the Cold War, where these practices were being mastered by both sides and causing privacy and security infiltrations within the United States. Specifically, KGB intelligence spies such as Alger Hiss and Julius and Ethel Rosenberg who were caught in New York City in 1950-51 as seen in Figure 1, were the start of hundreds of covert spy cases that would rise to the surface of the media during this time period [2]. With all of these spies rising to the forefront of the media, civilians were scared for their security and privacy within their own homes and everyday lives like never before. With government operatives becoming so skilled, technological developments were not yet necessary for privacy and security to become a growing concern for the American citizen.

Figure 1: Capture of Julius and Ethel Rosenberg, New York 1951 [2]

With this, the United States more so utilized the latter two intelligence practices to counter the Soviet Union, SIGINT and PHOTINT. Signals Intelligence, which was mainly composed of interception of radio and phone call signals, was an indispensable pillar of intelligence collection for the CIA. With the fears of espionage within the nation, it was paramount for the United States to have access to people's phone calls and radio communications, in order to protect the welfare of the nation as a whole. Similarly, photographic intelligence, using things such as cameras, reconnaissance aircraft, and satellites, were also used

as counter-intelligence practices in order to uncover KGB infiltration within the country. The FBI then established covert missions in the 1950's with the program established under the name "COINTELPRO", which aimed at disrupting organizations or individuals domestically which posed a threat to the current order. Although being ran by the FBI, "In 1986, a federal court determined that COINTELPRO was responsible for at least 204 burglaries by FBI agents, the use of 1,300 informants, the theft of 12,600 documents, 20,000 illegal wiretap days and 12,000 bug days" [3].

In this way, with the newfound fear in society regarding the Soviet Union, the US government took advantage of the fears of their own citizen's security, and used it to perform illegal activities to regulate domestic matters as well as international ones. As said by Rev. Jesse Jackson in a 1970's report, "When you have this feeling that the government really is watching you, you know, taps your telephone, maybe in your text files, it has a chilling effect. It takes away your freedom…it neutralizes people" [4]. Before the internet and the cyber-world even came to birth, the debate of privacy vs. security was already on the rise.

This led to the implementation of the Privacy Act of 1974, which set boundaries on the information the government could collect and distribute regarding individuals personal information. Similarly, in 1978, the Foreign Intelligence Surveillance Act was passed, "which basically put limits on secret national security wiretaps" [4]. With these newfound laws, the government began to finally take action among the imbalances between privacy and security regarding primitive intelligence practices, but the development of the cyber-world would bring much more debate to center stage.

Eventually, starting in 1969, the beginnings of the internet came to be, revolutionizing the way information was accessed, shared, and communicated globally. ARPANET, the first public packet-switched computer network, was the forerunner of the current day internet. ARPANET operated with the Network Control Program (NCP), which "would handle communication between hosts and could support the first commands, Telnet and File Transfer Protocol (FTP)" [5]. Interface Message Processor was also developed to pass messages between hosts. The first message over ARPANET happened on Oct. 29th, 1969 between UCLA and the Stanford Research Institute, permanently revolutionizing what was believed to be technology capable regarding the exchange of information.

Although ARPANET was mainly used for academic and research purposes, the internet would grow rapidly in the 1970's and this would come to quickly change. In 1977, Transmission Control Protocol / Internet Protocol (TCP/IP) was introduced and quickly replaced ARPANET's NCP in 1983. With this new protocol being implemented, computers on different networks communicate with one another through interconnected networks. This was the first time that computers were truly connected globally to an extent, making TCP/IP the founding protocol technology of the modern day internet. Civilian use of computer communication was rising at an

exponential rate, with the first emails, spam emails, list servers, and password and data encryption being developed here as civilians exchanged information online.

Even though the internet has begun its first steps, it was not yet popular enough with society for it to be a major privacy or security concern for the average citizen. This would all change in 1989, when the true internet was born by Tim Berners-Lee with his creation of the World Wide Web (WWW) [6]. This internet, the one that we used today, led to ARPANET being shut down only a year or so later, and the seamlessly communicating World Wide Web was now the sole operator among interconnected networks. With the implementation of the World Wide Web, it was now easy for anyone and everyone to exchange information along the internet, including businesses. Companies such as Ebay and Amazon were the first in 1994 to grab domain names and make websites for e-commerce [6], an example of which is seen in Figure 2. With business and products being sold over the internet, internet usage skyrocketed and quickly became commonplace in every household. With everybody using the internet, everyone in the world quickly became globally interconnected by exchanging information over the internet.

Figure 2: First registered trademark of Amazon Technologies, Inc. [6]

The connection of everybody over the internet quickly proved to result in security and privacy concerns. First of all, e-commerce websites and large corporations began to implement cookies on their websites, which would store small pieces of data on a user's computer and track their activities while they were on the browser. Geared with this information, corporations could easily practice targeted advertising; using what they know about individuals' activities in order to cater products that they would like. This was dangerous as it was so profitable, people's personal information quickly became a product of high demand, and this information would begin to be exchanged across organizations over the internet. With the elementary encryption and internet security that existed at the time, this introduced the first internet hackers; with dangerous ones being those who sought personal information in exchange for profit. Whereas computer viruses and worms used to only be able to get passed around with physical computer disks, "computer viruses and worms… could now grow with tremendous speed via the Internet" [7].

Lack of security on the internet was presenting privacy risks like never before seen; everyone's personal information was being easily streamlined and exchanged along the internet without many even realizing it. Quickly, society became concerned, leading to the government taking action. While security rose on the internet with DES and eventually AES being established by the government, privacy concerns were still of great concern. Government entities began to monitor people's activity on the internet like never before, both to capture hackers/criminals and to monitor suspicious activities and prevent data breaches. Although people were now much safer browsing on the internet than ever before, questions now rose

concerning just how much private information was being gathered by the government as opposed to petty criminals. People began to question whether or not the government was abusing their power; privacy concerns came to the forefront and security took a step back. This rise in privacy speculations would essentially disappear after the events of September 11th, 2001.

## The USA PATRIOT Act and Edward Snowden's Leaks

After the terrible events of the September 11th terrorist attacks, the public was scared of additional attacks and the U.S. government needed solutions that would ease their minds as well as help prevent further attacks. One of these was the USA PATRIOT Act. Signed into law by President George W. Bush on October 26th, 2001, the Patriot Act gave new powers to the federal government including expanding legal surveillance by law enforcement.

Previously there had been a couple different ways for intelligence agencies and law enforcement to obtain information. The use of FISA warrants were the main option to achieve electronic surveillance while a type of subpoena known as a National Security Letter (NSL) could be used to obtain business and financial records. Under Title V, electronic surveillance was expanded to the collection of emails and other internet metadata [7]. Among other things, changes were made to change the legal standard needed to use these methods from "'relevance' plus 'specific and articulable facts giving reason to believe' the target was a foreign power or an agent of one, to simple 'relevance' to an investigation to protect against international terrorism or clandestine intelligence activities' as described by M.J Woods [8]. It also eased the burden on intelligence and law enforcement agencies by changing the process of intelligence sharing between agencies to be much easier. This expansion of power created worry, but would not be realized until a few years later.

The Patriot Act contained sunset clauses that required it to be extended every few years. Specific provisions were modified and extended in each iteration. The USA PATRIOT Improvement and Reauthorization Act of 2005 included provisions that allowed for nondisclosure requirements meaning that companies that were subpoenaed using NSLs were required to keep silent about any requests for information and what information was released. The PATRIOT Sunset Extensions Act of 2011 extended several provisions. The USA Freedom Act was passed in 2015 modifying and extending several provisions. The Patriot Act finally expired in 2020 when it failed to be extended.

The full extent of how the Patriot Act was used was unknown until 2013 when an NSA subcontractor working for Booz-Allen Hamilton took action. Edward Snowden stole the most classified documents in history, an estimated value of over 1.7 million according to the pentagon [9]. He then released thousands of documents in June of 2013 to American journalists who proceeded to release and report on these leaks to the public. All this information painted a picture of how US agencies were abusing the Patriot act to spy on US citizens. One of these reports by Glenn Greenwald [10] describes how the Obama administration had used NSLs to collect

millions of customers' metadata from Verizon. Apparently this metadata does not require individual warrants as long as it does not contain the actual communications information, such as voice recordings or transcripts.Similar FISA requests had been increase in the past 5 years, with 21 approvals in 2009, 212 approvals by 2012, up to 564 orders approved between 2012 and 2014 [7].

A much larger leak, written by Greenwald and MacAskill, was the reveal of PRISM, a program run by the NSA that mass collects data from the 9 largest internet companies: Microsoft, Yahoo, Google, Facebook, PalTalk, YouTube, Skype, AOL, and Apple. However this data collection is different from the Verion leak as it includes “search history, the content of emails, file transfers, and live chats,” [11]. A lot of information about PRISM comes from an 11 slide presentation in the leak which was marked as Top Secret classification [12]. It details information about how the program works, from including a full list of providers and types of data collected, to how PRISM compares to another surveillance program called Upstream. PRISM directly mines information from the servers of large companies while Upstream collects communications on fiber cables and other communications infrastructure. It also included the collection and dataflow from PRISM and the tasking process for how government agencies would request for data collection and information using PRISM.

These FISA requests and use of the PRISM program were a cause of distress for privacy advocates as this collection of data was no longer contained to individuals suspected of terrorism, it now allowed for surveillance/ data collection on individuals that use the same cell tower or are within a similar internet community. While created by the NSA, PRISM was open to many agencies and was heavily used by the FBI and its National Threat Operations Center.

The president at the time of the Snowden Leaks, Barack Obama, was asked to respond to the leaks concerning his administration's use of FISA and NSLs as well as the PRISM program. This caused issues with his administration's reputations as president Obama had made many comments about his condemnation of the Patriot act and the issue of allowing for warrantless mass searches. When approached for comments, Obama defended the PRISM as well as his administration's use of FISA claiming that he believed they had come to a nice balance between Security and Privacy. He made a more general comment on the topic, stating "You can't have 100 percent security and also then have 100 percent privacy and zero inconvenience," Obama said. "We're going to have to make some choices as a society. ... There are trade-offs involved." [13].

## Legal Efforts

The Patriot Act in the aftermath of 911 was a major boost to national security by increasing surveillance capabilities, however, it was also a big encroachment on american citizens privacy. Initially set to expire due to “sunset” provisions, extended multiple times in 2005, 2006, and beyond in the form of other acts[14]. Major legal changes further increased

government authority came in the form of amendment to Foreign Intelligence Surveillance Act Section 702 in 2008. This act authorized the collection of foreign intelligence information from foreign nationals outside of the US[15]. This act was significant in the sense that the government gained the ability to conduct large scale surveillance without any need for warrant, and eventually allowed for programs like PRISM, which was revealed by Edward Snowden in 2013.

Next, in 2013 following Edward Snowden's reveal of mass surveillance by the government, and the public outcry there after. The USA Freedom Act was a response to reform the authorities of the government. Although it reauthorized some parts of the Patriot Act, it released new interpretations of the law that essentially ended bulk collection of telephone data by the NSA[16], while still allowing the agency access to telecommunications companies. This was a major step to swing the balance back in privacy rights, as the previous legal acts all were for national security at the expense of privacy. The surveillance practices continue to be modified afterwards to this day, in the ever changing environment of security vs privacy.

With the advent of modern mobile applications, an increasing amount of private data is stored online. The data breaches in the past few years have led to a heightened concern about privacy protections. One such incident, amongst many others, was the Facebook Cambridge Analytica leak. Cambridge Analytica was a data-analytics company that collected information from Facebook without consent and used the data to analyze then create targeted advertisements in an effort to swing election results in the US[17]. This incident happened as Cambridge Analytica took advantage of a loophole in the Facebook API system. This incident led to significant public backlash, legal scrutiny, and regulatory fines for Facebook. Following this, Facebook itself started implementing stricter data access policies.

As the flow of data becomes ever increasingly global, governments, companies and the public face an increasingly difficult task to protect their data when it's already in databases accessible in the cloud. Legal changes came after this period of massive data breaches in favor of consumer privacy rights protection by changing how data is collected.

Those concerns took the form of state statutes such as the California Consumer Privacy Act (CCPA). California modeled and implemented CCPA in 2018 after Europe's General Data Protection Regulation (GDPR), and is a major step towards enhancing privacy and rights in that area of consumers[18]. The consumers gain the following rights: know what personal data is being collected, whether that data is being sold, decline the sale of that data, request deletion of that data, and not to be discriminated against for exercising those rights. Following California, other states are implementing their own versions of similar legal acts to help protect privacy.

From the perspective of national security vs privacy protection, technology advancements, including the internet and more recently Artificial Intelligence, led to the amount of data present on the internet cloud increasing exponentially where security of data can no longer be guaranteed has led to new legislation aimed at changing how data is collected. The

role, and to an extent, business model of companies drastically changes in face of more strict regulation on consumer privacy.

## Debate of Security vs Privacy

As laws like the Patriot Act begin to infringe on public privacy to enhance security, the debate over privacy and security begins to emerge. Unfortunately, full security and full privacy cannot co-exist together. In order to have one, the other must be abridged. With that being said, what should be favored security or privacy?

First let's begin with the case for security. The reason laws like the Patriot Act were created were a result of the aftermath of devastating events like the attacks on September 11th, 2001. The goal is to stop terrible acts like terrorism to ensure public safety and national security. Many people, including government agencies, argue that security via cyber surveillance is absolutely necessary for this. It's also argued that more security is necessary to keep up with trends and threats in the cyber security world. On top of this is another point which is that having quick access to personal data can be beneficial in times of crisis like a natural disaster or a health emergency. In short, security helps to ensure more safety and well-being.

Next up, let's move onto the case for privacy. When laws like the Patriot Act were created, privacy was infringed on. The US government would now legally be able to listen in on communications over the internet. The implications of this are going to affect autonomy, civil liberties, and human rights. Autonomy is the ability to self-govern, if people are aware of them being monitored then they will behave differently. Civil liberties are rights guaranteed by the government and violating this can break trust between the public and government institutions. Human rights are the most philosophical part of this case, but many argue that privacy is a human right. Supporters of privacy argue that all of these are violated by laws like the Patriot Act. They argue that the law isn't effective enough, so it doesn't outweigh its downsides. In short, privacy ensures more freedom.

Both arguments for privacy and security are very reasonable and a clear answer doesn't present itself. The debate is undoubtedly very subjective. With this being said, where does the public/American population stand on this debate? The answer to this varies and has a lot to do with recency bias. A study done by the Pew Research Center found that after the San Bernardino and Paris shootings, 56% of Americans favored security and more government anti-terror policies to 28% concerned that policies have gone too far with respect to civil liberties, whereas after the leakage of the Edward Snowden case 47% of Americans believe that civil liberties policies had gone too to a 35% saying there wasn't enough security/anti terror policies [19]. This study appears to show that the public isn't quite sure of which option is better either. In order to gain a stronger intuition towards finding the correct direction to take, a more in depth analysis is needed here.

## Public Perception

In a world defined by our technological evolution and increased connectivity, we must constantly adapt to varying levels of privacy and security in our lives. As discussed above, governments collect data about its citizens. Some collection is necessary to help protect the safety of society; however, many feel that they have overstepped and are invading its citizens' privacy. Companies have also started reaching into our personal lives more and more. It is up to us as users to determine if they are willing to sacrifice some privacy for the product or service that these companies provide. Lastly, there are a couple ways to help protect ourselves, but the best way to protect yourself from invasive companies is to completely stop using their products or services.

A survey by the Pew Research Center found that "Americans are largely concerned and confused about how their data is being used" [20]. This is for both companies and the government. They found that adults are concerned about how companies (81%) and the government (71%) use data they collect and have little to no understanding about what companies (67%) or the government (77%) does with the data they collect. A similar survey also found that there has been an overall increase in concern by 7% from 2019 to 2023 about how the government uses the data, but republicans are particularly concerned with an increase of 14% during the same time. These statistics highlight that companies and the government are not doing enough to explain what information they collect, and there is not enough transparency with what happens after it is collected.

Many people will openly share personal moments from their lives on social media, but the vast majority do not trust social media companies or their leaders. The same survey by the Pew Research Center found that 77% of U.S. adults report having very little or no trust that leaders will publicly admit to their mistakes and take responsibility when they misuse or compromise users' personal data. Similarly, 76% don't trust them to not sell users' personal data to others without consent. And a staggering 71% don't trust their own government to hold social media companies accountable if they misuse or compromise users' personal data. Surprisingly, despite having little trust in the government, 72% of Americans would support more regulation for all companies in regards to data they collect and what they can do with it. There is also a growing number, 72% from 63% in 2019, who have little or no understanding of data privacy laws.

Privacy policies are quickly accepted, but Americans rarely take the time to read them. 56% report having always or often accepting privacy policies without reading them. And only 7% find them effective at sharing how their data is used. One reason for this is that privacy policies are written in legal writing and can be hard for some to read and understand without some secondary education. A trend highlighted by the author was that increased levels of education resulted in decreased belief in effectiveness for privacy policies. 63% with some education found the policies to be not effective, where only 46% of high school or less educated

individuals felt the same way. Depressingly, 69% of Americans don't care about privacy policies and only view them as a step to get past before using a product or service. And this varies with education level.

One good note is that Americans generally trust themselves to make the right decisions and many take steps to help their privacy on the web. However, we're skeptical that even when we do the right thing that companies will actually do what they say they will to protect our security and privacy. The survey found that Americans are nearly split on what to prioritize in a password: a long and secure one or a memorable one. Due to the sheer number of passwords we do need to remember, nearly 70% feel overwhelmed. To help, some Americans (32%) reported using a password manager.

There are many steps that people can take to help protect their online privacy, but none of them are perfect solutions. One of the simplest options is to install a trusted content blocker like uBlock Origin or Adguard. These block things like trackers, ads, and malware that companies use to get into a user's personal life and impact some of their decisions. Next, users can switch the search engine they use on a daily basis. Options like DuckDuckGo and Startpage are better for privacy and don't serve ads at the top of the results like Google or Bing. This is possible regardless of browser but some may try to hide this setting from the user. Then, if these are not enough, users can switch their browser to options like Firefox or Brave who focus on privacy. Lastly, users can decide to not use products or services if they have a history of disrespecting users' data. Peer pressure can make it hard to stop using these services so users must decide if they value their privacy or digital relationships more.

## Conclusion and Takeaways

Tracing roots all the way to protecting the US during the Cold War to man made disasters like terrorism today, has exposed the need for security. In an attempt to mitigate these events to ensure national security and public safety, the US government has created several laws such as the Patriot Act to increase surveillance over public communications. While surveillance of individuals and groups has long since been common practice, digital surveillance creates this larger issue of mass data collection. This has led to large-scale worries surrounding how the US government monitors its citizens' data and communications. It is a very difficult position for a government to balance the privacy of their citizens with the security of their country.

Before the Patriot Act, the United States experienced a great amount of concerns regarding privacy and security during the Cold War. This led to the development of laws such as the Privacy Act of 1974 and the Foreign Intelligence Surveillance Act in an attempt to address these imbalances. The emergence of the internet in the late 20th century brought about a new era, where concerns expanded beyond primitive intelligence practices to encompass what is now the sole concern of the modern day world: online privacy and security.

The events of September 11, 2001, marked a turning point which granted more security powers to the government. However, revelations from Edward Snowden exposed the extent to which the government was watching its citizens, creating public outcry and raising questions about the trade-offs between security and privacy and what we should desire as citizens.

Legal efforts, such as the USA Freedom Act, aimed to strike a balance between national security and privacy rights. As technology continued to advance, data breaches and privacy concerns prompted legislative changes, including state statutes like the California Consumer Privacy Act. Public perception today reflects an expressing worry about how companies and the government use their data, but also a concerning lack of knowledge of just how many places their private information is available on the internet. Privacy policies are often accepted without reading, and although there is a growing desire for more understanding and control over personal data, there isn't much people actually do about it. Individuals can and should begin to take steps in order to protect their online privacy, such as using content blockers, alternative search engines, and privacy-focused browsers. However, it is also true that these precautions are not infallible; illustrating the continuous difficulties in striking a balance between security and privacy, as well as the necessity for citizens to be always aware of their ever-evolving privacy rights and granted accessibilities.

All in all, the interplay between privacy and security is one that will always be clashing together so long as society lives alongside the internet. In order to minimize the collateral that comes with each, citizens must be more informed and do their best to practice safe and private actions while using the internet and always advocate for their privacy and security rights. As it can be seen, technology will only become exponentially greater and greater as the years go on, requiring us to be just as much more cautious and safe when utilizing the beautiful technological feats we create.

## References

[1] M. Sulick, "Guide to Intelligence in the Cold War," *INTELLIGENCE: Journal of the International Association for Intelligence Education*, vol. 21, no. 1, Winter 2014-15. [Online].https://www.afio.com/publications/SULICK_Michael_Guide_to_Intelligence_in_the_Cold_War_from_INTEL_WINTER2014-15_Vol21_No1.pdf

[2] Klehr, Harvey. "Cold War Spy Cases," *Bill of Rights Institute*. [Online]. https://billofrightsinstitute.org/essays/cold-war-spy-cases

[3] Murphy, Laura W. "Trust Us, We're from the Government: ACLU Looks at Domestic Surveillance and the Need to Watch the Watchers," ACLU Press Releases. [Online]. https://www.aclu.org/press-releases/trust-us-were-government-aclu-looks-domestic-surveillance-and-need-watch-watchers#:~:text=In%201986%2C%20a%20federal%20court,days%20and%2012%2C000%20bug%20days

[4] Gordon, Ed. "COINTELPRO and the History of Domestic Spying", *NPR*. [Online]. https://www.npr.org/templates/story/story.php?storyId=5161811

[5] Wright, Gavin. "ARPANET," *TechTarget*. [Online]. https://www.techtarget.com/searchnetworking/definition/ARPANET#:~:text=The%20U.S.%20Advanced%20Research%20Projects,for%20academic%20and%20research%20purposes

[6] Bourgeois, David T. "World Wide Web," *Open Textbooks*. [Online]. https://www.opentextbooks.org.hk/ditatopic/18343#:~:text=In%201989%2C%20Tim%20Berners%2DLee,to%20share%20information%20about%20themselves

[7] S. Ackerman, "Snowden disclosures helped reduce use of Patriot Act provision to acquire email records," *the Guardian*, Sep. 29, 2016. Accessed: Dec. 05, 2023. [Online]. https://www.theguardian.com/us-news/2016/sep/29/edward-snowden-disclosures-patriot-act-fisa-court

[8] M. J. Woods, "Counterintelligence and Access to Transactional Records: A Practical History of USA PATRIOT Act Section 215," *JOURNAL OF NATIONAL SECURITY LAW & POLICY*, vol. 1, no. 37, pp. 37–71, 2005.

[9] C. Strohm and D. Q. Wilber, "Pentagon Says Snowden Took Most U.S. Secrets Ever: Rogers," *Bloomberg.com*, Jan. 10, 2014. Accessed: Dec. 05, 2023. [Online]. https://www.bloomberg.com/news/articles/2014-01-09/pentagon-finds-snowden-took-1-7-million-files-rogers-says

[10] G. Greenwald, “NSA Collecting Phone Records of Millions of Verizon Customers Daily,” *The Guardian*, Jun. 06, 2013. Accessed: Dec. 05, 2023. [Online]. https://www.theguardian.com/world/2013/jun/06/nsa-phone-records-verizon-court-order

[11] G. Greenwald and E. MacAskill, “NSA Prism program taps in to user data of Apple, Google and others,” *The Guardian*, Jun. 07, 2013. Accessed: Dec. 05, 2023. [Online]. https://www.theguardian.com/world/2013/jun/06/us-tech-giants-nsa-data

[12] “NSA Prism program slides,” the Guardian, Nov. 2013. Accessed: Dec. 05, 2023. [Online]. https://www.theguardian.com/world/interactive/2013/nov/01/prism-slides-nsa-document

[13] M. Spetalnick and S. Holland, “Obama defends surveillance effort as ‘trade-off’ for security,” *Reuters*, Jun. 07, 2013. Accessed: Dec. 05, 2023. [Online]. https://www.reuters.com/article/idUSBRE9560VB/

[14] U.S. Department of Justice, "Legal Information Institute," U.S. Department of Justice, [Online].. Dec. 6, 2023.

https://www.justice.gov/archive/ll/archive.htm

[15] "Section 702 Basics Infographic," Office of the Director of National Intelligence, [Online]. Dec. 6, 2023.

https://www.dni.gov/files/icotr/Section702-Basics-Infographic.pdf.

[16] J. Sensenbrenner and P. Leahy"USA Freedom Act of 2015" U.S. Congress. [Online]. Dec. 6, 2023.

https://www.congress.gov/114/plaws/publ23/PLAW-114publ23.htm.

[17] N. Confessore, "Cambridge Analytica Scandal Fallout," The New York Times, Apr. 4, 2018. [Online]. Dec. 6, 2023. https://www.nytimes.com/2018/04/04/us/politics/cambridge-analytica-scandal-fallout.html.

[18]R. Bonta, "California Consumer Privacy Act (CCPA)," Office of the Attorney General, State of California Department of Justice, [Online]. Dec. 6, 2023.

https://oag.ca.gov/privacy/ccpa

[19] Maniam, Shiva. “Americans Feel the Tensions between Privacy and Security Concerns.” *Pew Research Center*, Pew Research Center, 19 Feb. 2016, www.pewresearch.org/short-reads/2016/02/19/americans-feel-the-tensions-between-privacy-and-security-concerns/.

[20] C. McClain, et al. "How Americans View Data Privacy," Pew Research Center, 18 Oct. 2023. [Online]. https://www.pewresearch.org/internet/2023/10/18/how-americans-view-data-privacy/